

Light and Atomic Spectra

Answer key with worked solutions

A. Waves and photons

1. Order of energy.

From lowest to highest energy: **radio waves** → **microwaves** → **infrared** → **visible** → **ultraviolet** → **X-rays**

2. Microwaves.

(a) 12.2 cm = 0.122 m

$$\nu = \frac{c}{\lambda} = \frac{2.998 \times 10^8 \text{ m/s}}{0.122 \text{ m}} \\ = 2.46 \times 10^9 \text{ s}^{-1}$$

(b)

$$E = h\nu = 6.626 \times 10^{-34} \text{ J s} \times 2.457 \times 10^9 \text{ s}^{-1} \\ = 1.63 \times 10^{-24} \text{ J}$$

3. Blue light.

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J s} \times 2.998 \times 10^8 \text{ m/s}}{450 \times 10^{-9} \text{ m}} \\ = 4.41 \times 10^{-19} \text{ J}$$

Per mole: $4.414 \times 10^{-19} \text{ J} \times 6.022 \times 10^{23} \text{ mol}^{-1} = 2.66 \times 10^5 \text{ J/mol} = 266 \text{ kJ/mol}$

4. UV and bond breaking.

(a)

$$E = \frac{6.626 \times 10^{-34} \text{ J s} \times 2.998 \times 10^8 \text{ m/s}}{254 \times 10^{-9} \text{ m}} \\ = 7.82 \times 10^{-19} \text{ J}$$

(b) Energy per bond:

$$E = \frac{498 \times 10^3 \text{ J/mol}}{6.022 \times 10^{23} \text{ mol}^{-1}} = 8.270 \times 10^{-19} \text{ J} \\ \lambda = \frac{hc}{E} = \frac{6.626 \times 10^{-34} \text{ J s} \times 2.998 \times 10^8 \text{ m/s}}{8.270 \times 10^{-19} \text{ J}} \\ = 2.40 \times 10^{-7} \text{ m} = \mathbf{240 \text{ nm}}$$

Longer wavelengths carry less energy per photon, so 240 nm is the longest that works.
(The 254 nm lamp is just too weak to break O=O.)

B. Line spectra and the Bohr model

5. Spectra.

A **continuous spectrum** contains all wavelengths (a hot solid). A **line spectrum** contains only certain exact wavelengths (an excited gas). Each line is a photon with energy equal to the gap between two levels; since only certain wavelengths appear, only certain energies are allowed: the levels are **quantized**.

6. Bohr calculations.

(a)

$$\Delta E = 2.179 \times 10^{-18} \text{ J} \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = 4.086 \times 10^{-19} \text{ J} \\ \lambda = \frac{6.626 \times 10^{-34} \text{ J s} \times 2.998 \times 10^8 \text{ m/s}}{4.086 \times 10^{-19} \text{ J}} \\ = 4.86 \times 10^{-7} \text{ m} = \mathbf{486 \text{ nm}} \text{ (visible, blue-green)}$$

(b)

$$\Delta E = 2.179 \times 10^{-18} \text{ J} \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = 1.634 \times 10^{-18} \text{ J} \\ \lambda = \frac{6.626 \times 10^{-34} \text{ J s} \times 2.998 \times 10^8 \text{ m/s}}{1.634 \times 10^{-18} \text{ J}} \\ = 1.22 \times 10^{-7} \text{ m} = \mathbf{122 \text{ nm}} \text{ (ultraviolet)}$$

7. Ionization from $n = 2$.

$E_2 = -\frac{2.179 \times 10^{-18} \text{ J}}{2^2} = -5.448 \times 10^{-19} \text{ J}$, so the energy needed is $5.45 \times 10^{-19} \text{ J}$ per atom.

Per mole: $5.448 \times 10^{-19} \text{ J} \times 6.022 \times 10^{23} \text{ mol}^{-1} = 3.28 \times 10^5 \text{ J/mol} = \mathbf{328 \text{ kJ/mol}}$

C. Challenge

8. One photon or one mole?

182 kJ is the energy of a **mole** of photons (6.022×10^{23} photons), not of one photon. For one photon:

$$\begin{aligned} E &= \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J s} \times 2.998 \times 10^8 \text{ m/s}}{656 \times 10^{-9} \text{ m}} \\ &= \mathbf{3.03 \times 10^{-19} \text{ J}} \end{aligned}$$